



CREATING AI LEADERS



BIRMINGHAM CITY
University

Title: Predicting Airline Passenger Satisfaction Using Machine Learning Algorithms

Name: Suprem Shrestha

Student ID: 25123849

Module Title: Artificial Intelligence &
Machine Learning

Module code: CMP6202

Date: 25th January 2026

Abstract

The aviation sector is very competitive and hence passenger satisfaction is key to profitability and loyalty. In this project, a machine learning pipeline is built to make predictions of airline passenger satisfaction based on demographic, flight-related and service-quality passenger characteristics based on 129888 records of passengers. The methodology consists of exploring data analysis, preprocessing and ensemble classification based on Random Forest and LightGBM. Findings indicate that the ensemble approaches are very effective with the best results being with a Random Forest (accuracy = 93.15%, F1 = 0.9315) followed by LightGBM (accuracy = 91.95%, F1 = 0.91) and a baseline of a Logistic regression (accuracy = 87%, F1 = 0.872). There is a significant emphasis on digital services and comfort with Online Boarding, In-flight Wi-Fi, and Class as the main contributors to satisfaction. These results imply that airlines will have opportunities to invest in digital infrastructure and high-quality services based on predictive data to enhance customer retention.

Keywords: Airline Passenger Satisfaction; Machine Learning; Random Forest; Ensemble Learning; Exploratory Data Analysis; Feature Importance; Class Imbalance; SMOTE; Service Quality

Table of Contents

Abstract.....	2
Table of Contents.....	3
Table of Figures.....	5
Figure 1: Distribution of Target Variable (Passenger Satisfaction).....	5
Abbreviations.....	6
1. Report Introduction.....	7
1.1. Research Gaps.....	7
1.2. Objectives.....	7
2. Dataset Description.....	8
2.1. Dataset Introduction.....	8
2.2. Machine Learning Type Identification.....	8
3. Exploratory Data Analysis (EDA).....	8
3.1. Operations.....	9
3.1.1. Data Inspection:.....	9
3.1.2. Descriptive Statistics:.....	9
3.1.3. Univariate Analysis.....	9
3.1.4. Bivariate Analysis.....	11
3.1.5. Correlation Analysis.....	12
3.2. Conclusion from EDA.....	13
4. Experiment Design.....	13
4.1. Justification of Model.....	13
4.1.1. Logistic Regression (LR).....	13
4.1.2. Random Forest (RF).....	13
4.1.3. LightGBM (Gradient Boosting).....	14
5. Data Cleaning and Preprocessing.....	14
5.1. Handling Missing Values.....	14
5.2. Outlier Treatment.....	14
5.3. Feature Engineering.....	15
5.4. Categorical Encoding.....	15
5.5. Handling Class Imbalance (SMOTE).....	15
5.6. Data Quality Assessment.....	15
6. Model Development.....	16
6.1. Train-Test Split Strategy.....	16
6.2. Architecture and Configuration.....	16
6.2.1. Logistic Regression (Type: GLM):.....	16
6.2.2. Random Forest (Type: Bagging Ensemble):.....	16

6.2.3. LightGBM (Type: Gradient Boosting Ensemble).....	17
7. Evaluation Metrics & Results.....	17
7.1. Comparative Performance Analysis.....	17
7.2. Confusion Matrix Analysis.....	18
7.3. Feature Importance.....	19
7.4. Permutation Importance Analysis.....	19
7.5. Learning Curve Analysis.....	20
7.6. Interpretation of Results.....	21
8. Conclusion.....	21
8.1. Summary.....	21
8.2. Limitations.....	22
8.3. Reflection on Individual Learning.....	22
8.4. Future Recommendations.....	22
References.....	23

Table of Figures

Figure 1: Distribution of Target Variable (Passenger Satisfaction)

Figure 2: Age Distribution

Figure 3: Flight Distance Distribution

Figure 4: Delay Distributions

Figure 5: Satisfaction by Class

Figure 6: Satisfaction by Type of Travel

Figure 7: Service Ratings by Satisfaction Group

Figure 8: Correlation Heatmap

Figure 9: Outlier Before Treatment

Figure 10: Outlier After Treatment

Figure 11: Model Performance Analysis

Figure 12: Confusion Matrix

Figure 13: Feature Importance

Figure 14: Permutation Importance

Figure 15: Learning Curve

Abbreviations

Abbreviation	Full Form
AQR	Airline Quality Rating
AUC	Area Under the Curve
EDA	Exploratory Data Analysis
F1	F1-Score (Harmonic mean of precision and recall)
GOSS	Gradient-based One-Side Sampling
IQR	Interquartile Range
LightGBM	Light Gradient Boosting Machine
LR	Logistic Regression
MAE	Mean Absolute Error
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine

1. Report Introduction

Tough competition, low-profit margins and customer demands represent the international aviation industry. The satisfaction of passengers is a major indicator of loyalty to an airline and its stable revenues and competitive edges (Zeithaml et al., 1996; Hess and Polak, 2005). Non-linear relationships such as satisfaction are, however, a result of numerous interacting factors (service ratings, delays, travel purpose, and cabin class), and it is not easy to capture such relationships using traditional analysis methods (Park et al., 2004). Machine learning can be an orderly approach to learn patterns based on large-scale passenger data and generate predictive models that can be used to facilitate improvement in operations, as well as on a strategic level, in the area of service (Provost and Fawcett, 2013).

1.1. Research Gaps

Although previous studies looked at the quality of airline services, there are significant gaps concerning the application of sophisticated ensemble machine learning techniques (Park et al., 2004; Saha and Theingi, 2009). Numerous studies use linear methods of analysis including structural equation modeling or logistic regression, which do not perform well in understanding non-linear relationships between service factors (Hastie et al., 2009). Moreover, few studies have compared bagging and boosting methods to large, class balanced aviation data. Current models are claimed to have accuracy of close to 86 with a few reaching up to 90 with optimization ensembles such as LightGBM (Saha and Theingi, 2009). The project fills these gaps by considering high-performance ensemble classifiers and correcting the existence of imbalance in classes through SMOTE (Chawla et al., 2002).

1.2. Objectives

The main goal of this report is to create a very accurate predictive model on airline passenger satisfaction classification.

- Compare Logistic Regression, Random Forest and LightGBM to decide which gives the best compromise between the accuracy, interpretability and low computational costs.
- Improve model validity and generalization across passenger segments applying IQR-based outlier capping and SMOTE oversampling.
- Visualize the ranking of the 14 service attributes and their contribution to the satisfaction using feature important analysis.
- Provide practical insights that allow airlines to move towards proactive instead of reactive management of satisfaction.

2. Dataset Description

2.1. Dataset Introduction

The data set in the study is a dataset of a survey of the satisfaction of passengers of the airlines on Kaggle, which has 129,880 records and 24 attributes. It covers demographic information, flights type, and rating of quality services via a 1- 5 scale. The binary target variable, Satisfaction, categorizes the responses based on being satisfied or neutral/dissatisfied with minor imbalance in classes. The dataset is realistic with skewed delay variables, outliers and few missing values, which are suitable to analyse in classification (Hastie et al., 2009).

- **Feature Categories:**
 - **Demographic:** Age, Gender, Customer Type
 - **Travel Attributes:** Type of Travel, Class, Flight Distance
 - **Operational Metrics:** Departure Delay, Arrival Delay
 - **Service Quality (14 attributes):** Seat Comfort, Food and Drink, In-flight, Entertainment, Wi-Fi, Boarding Process, Gate Location, Baggage Handling, Check-in Service, In-flight Service, Cleanliness, Online Boarding, and others
- **Target:** satisfaction (Binary: 'Satisfied' or 'Neutral/Dissatisfied').

2.2. Machine Learning Type Identification

The present project uses controlled machine learning in a binary classification problem because the data has labeled data in which the target variable (Satisfaction) is known directly and used to train the data. The use of classification algorithms is best suited to the given scenario, as such algorithms define boundaries of the decision depending on the feature patterns.

Three models have been chosen as they are representative of various philosophies: a simple linear risk, which means that a logistic regression is used as a baseline to guarantee the simplicity of the model and improve the efficiency of the proposed algorithm; a bagging mechanism, which is an ensemble of multiple decision trees, thus it once again addresses non-linearity and mixed-type data efficiently (Breiman, 2001); and LightGBM, an updated gradient boosting framework with the facility to be fast and accurate on large tabular data through histogram-based splitting and leaf-wise growth (Ke et al., 2017).

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is used for data quality inspection, study distributions, and understanding relationships between features and the target (Tukey, 1977). EDA is a conventional practice of recognising problems and guiding modelling decision making prior to training (Hastie et al., 2009).

3.1. Operations

3.1.1. Data Inspection:

The first step was analyzing sample records and confirming the type of data in each column. The data were both categorical and numeric, so various preprocessing strategies were to be used. None of them had duplicate records. There were only a few missing values, which were concentrated mostly on the Arrival Delay feature making data cleaning easier.

3.1.2. Descriptive Statistics:

Numerical characteristics (mean, median, quartiles) were transformed into summary statistics and frequency counts were calculated in categorical variables. This showed large deviations in Flight Distance and delay-related characteristics, having numerous zero values, and few very large values. These observations indicated skewness and skewness outliers that needed some analysis.

3.1.3. Univariate Analysis

The target analysis indicated the presence of moderate level class imbalance with about 57% of the passengers not satisfied or Neutral, and 43% of those satisfied.

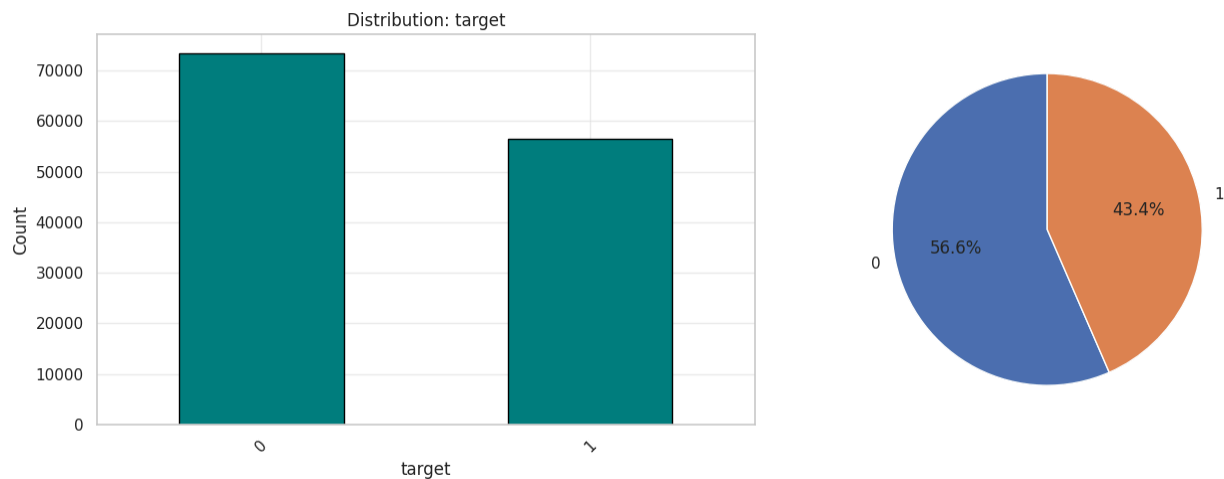


Figure 1: Distribution of Target Variable (Passenger Satisfaction)

Observation: This imbalance is not extreme. However, this suggests the use of SMOTE later on for removal of bias in the majority class when the model is being trained (Chawla et al., 2002).

In the case of numerical features, a histogram and a Kernel Density Estimates (KDE) were used to study the distributions.

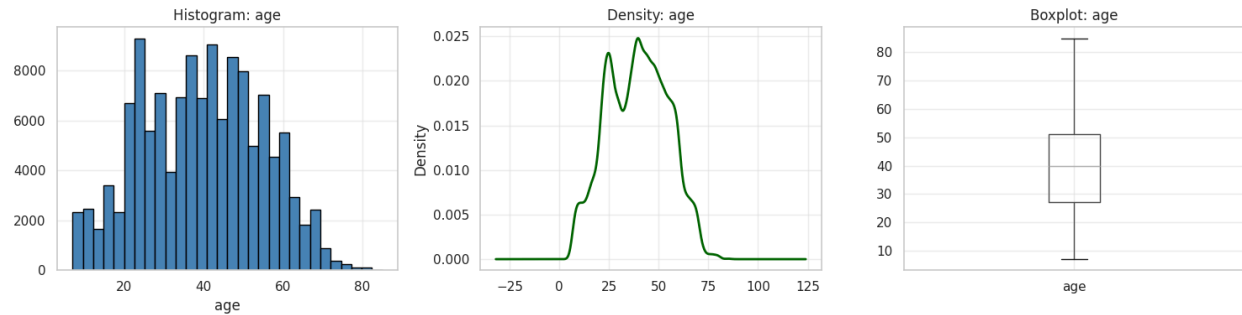


Figure 2: Age Distribution

Observation: The age distribution is approximately bellshaped and varies slightly through the middle age groups. This indicates the different traveling habits of people across various age groups.

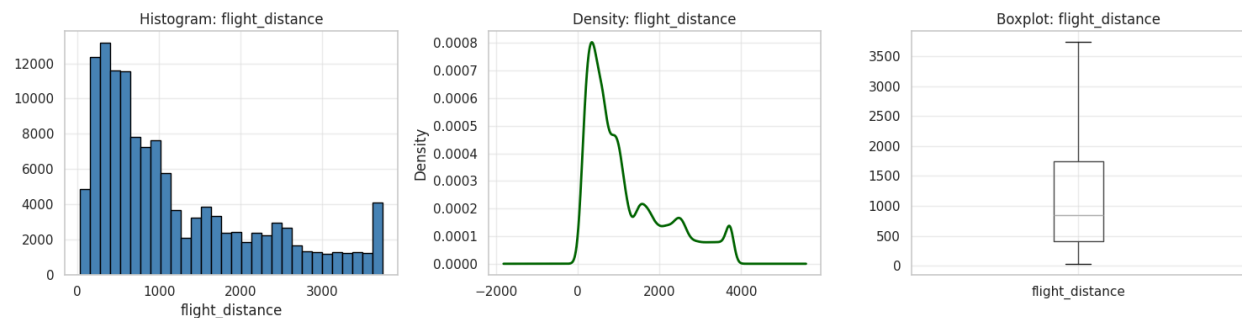


Figure 3: Flight Distance Distribution

Observation: The Flight Distance is heavily skewed to the right. The long-haul journeys are very few, and the short-haul are the majority. This skew creates a possibility of non linear distance related effects.

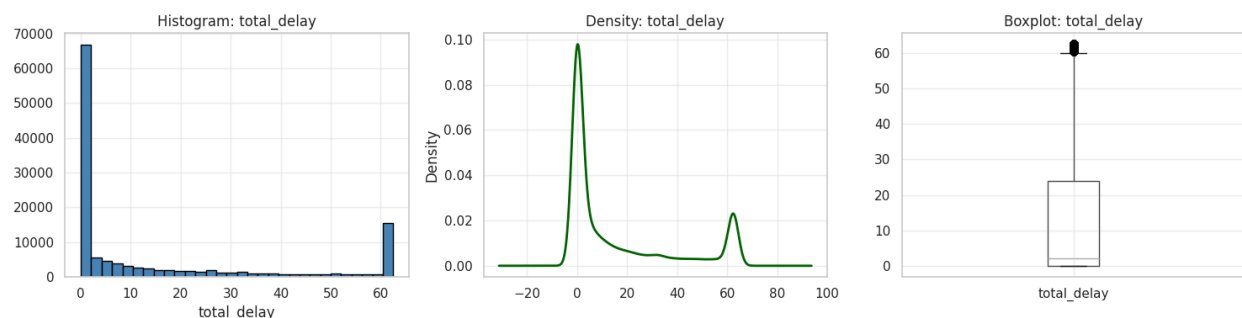


Figure 4: Delay Distributions

Observation: Delay variables consist of extreme values proving outlier presence. These extreme delays may not be uniform in affecting the learning of models when ignored. So, vulnerable preprocessing should be done.

3.1.4. Bivariate Analysis

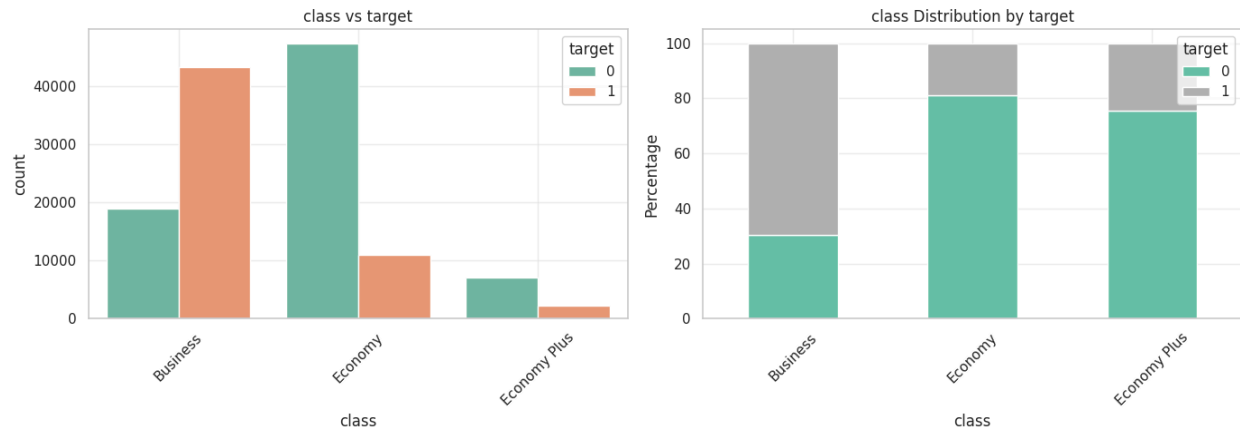


Figure 5: Satisfaction by Class

Observation: The level of satisfaction of Business Class passengers is much larger than that of Economy and Economy Plus passengers due to differences in services and comfort.

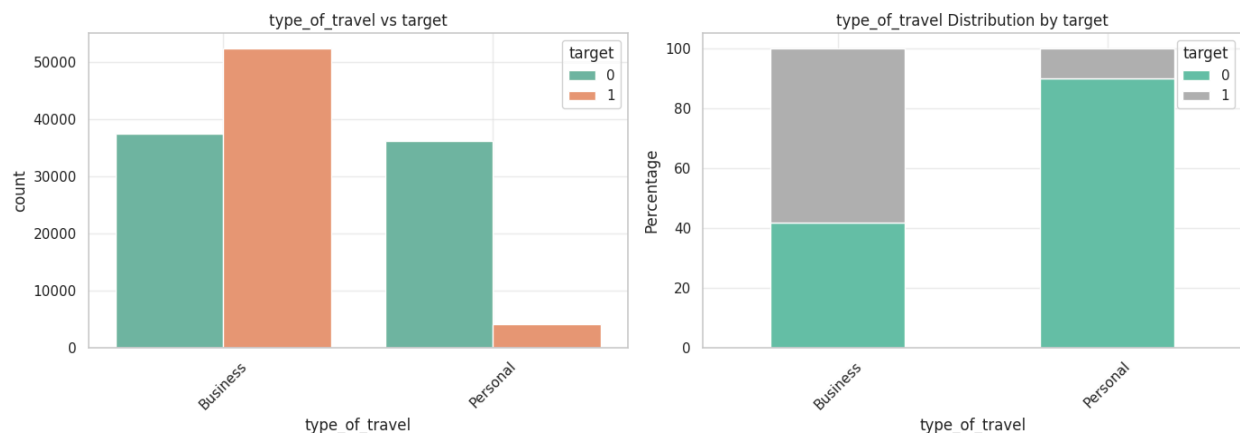


Figure 6: Satisfaction by Type of Travel

Observation: Business travelers are more likely to be satisfied than personal travelers, probably because of corporate benefits, priority services, or have a higher expectation level.

Group comparisons were also checked (mean/median rating by satisfaction group) in the case of ordinal service rating (1-5).

Observation: Some of the service attributes, especially, Online Boarding and In-flight WiFi Service have the highest deviations between the satisfied passengers and dissatisfied passengers, showing they are strong in distinguishing passengers experience.



Figure 7: Service Ratings by Satisfaction Group

3.1.5. Correlation Analysis

To determine relationships between numerical features and determine multicollinearity, a correlation heatmap was created.

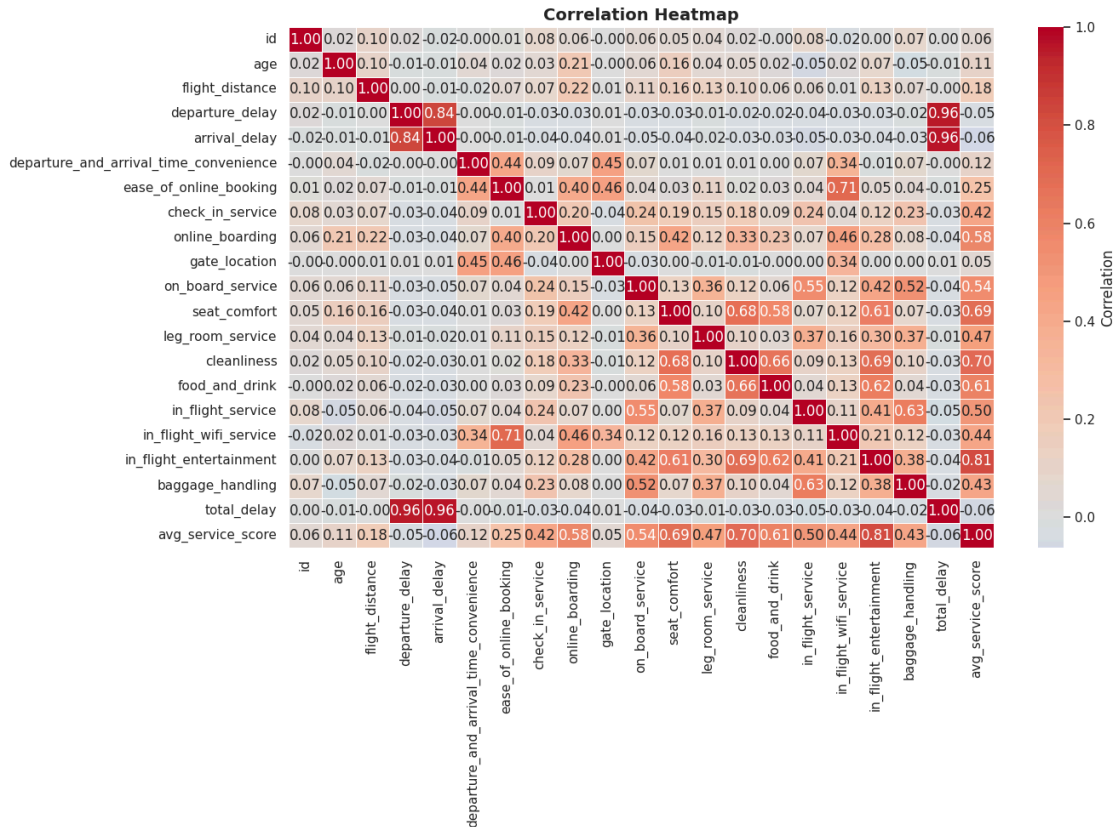


Figure 8: Correlation Heatmap

- **Observation:** The correlation between Departure Delay and Arrival Delay was very strong (≈ 0.96), which indicates multicollinearity. This redundancy may have an adverse effect on certain models since both variables may be almost translating the same information (Dormann et al., 2013). To overcome this, a Total Delay was developed, and tree based models were used and left to progressively manage correlated inputs.

3.2. Conclusion from EDA

The EDA process came up with three main conclusions:

- **Data Quality:** Missing data were insignificant and limited to Arrival Delay which had to be imputed. Features of delay had very large outliers and it was desirable that a robust handling be employed to avoid skewing of models.
- **Key Drivers:** Service quality attributes, namely Online Boarding and In-flight WiFi demonstrate the greatest difference in satisfaction classes, meaning they play a crucial role in the passenger experience.
- **Non-Linearity:** The correlation between the service rating (ordinal) and satisfaction is not always linear; ordinal service rating of 4-5 is disproportionately more likely to lead satisfaction. This is in favor of using tree models that are non-linear (Breiman, 2001).

4. Experiment Design

4.1. Justification of Model

4.1.1. Logistic Regression (LR)

Logistic Regression was chosen as the baseline model because it is easy to use, interpret and is common in conducting research on the service quality. Being a Generalized Linear Model, it has some explicit coefficient estimates which describe the effect of individual features on the likelihood of passenger satisfaction. It is computationally inexpensive and can thus be used in real-time prediction and benchmarking more complex models (Hosmer et al., 2013).

4.1.2. Random Forest (RF)

Random Forest was selected since it is a powerful non-Linear ensemble model that can model complex interactions of features without requiring a lot of feature engineering. Bootstrap aggregation helps RF to reduce variance and is also resistant to noise and outliers that are quite common among survey data (Breiman, 2001). It also gives feature importance scores, which have moderate interpretability.

4.1.3. LightGBM (Gradient Boosting)

LightGBM was chosen as a modern gradient boosting model that is optimized to work on large-scale tabular data based on leaf-wise tree growth. It utilizes Gradient-based One-Side Sampling and feature bundling to train faster, the method providing high-quality predictive accuracy. This renders LightGBM especially appropriate to model convoluted and non-linear satisfaction patterns on high volumes of data (Ke et al., 2017).

5. Data Cleaning and Preprocessing

5.1. Handling Missing Values.

There were 393 data points (around 0.3% of the data) missing in the Arrival Delay feature. To account for these missing values the median imputation was used in place of the mean imputation because the delay data is heavily skewed (Hastie et al., 2009). The extreme delay values may greatly distort the mean and the median represent a better and stronger estimate. This method guaranteed a small amount of information loss and at the same time maintained the statistical integrity of the feature.

5.2. Outlier Treatment

A number of the numerical characteristics, especially Flight Distance and the variables associated with delays had large numbers of outliers with delays of up to 1,000 minutes. The Interquartile Range (IQR) capping method was used to alleviate their effect (Tukey, 1977). The values which fell out of the 1.5X IQR were adjusted to the corresponding endpoints, the rest of the observations were retained to avoid having extreme values which influence the training of the model disproportionately.

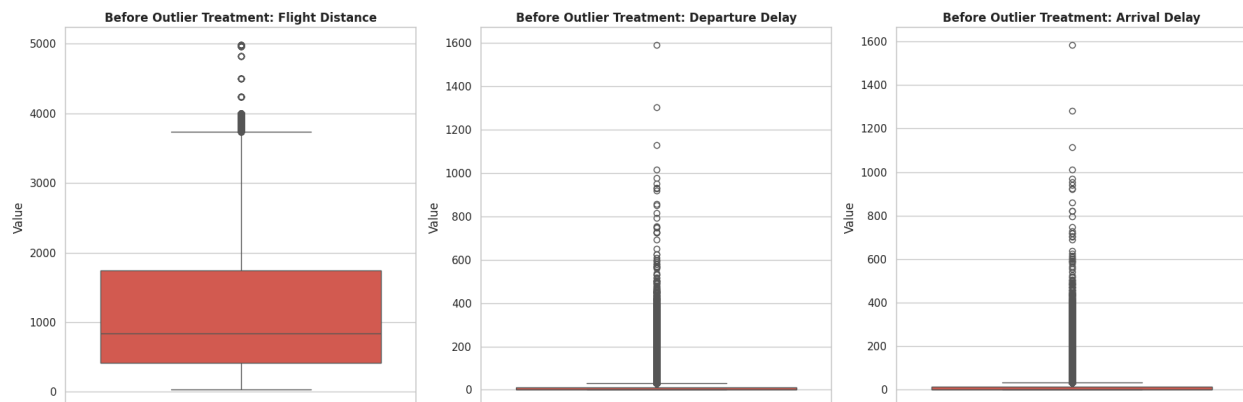


Figure 9: Outlier Before Treatment

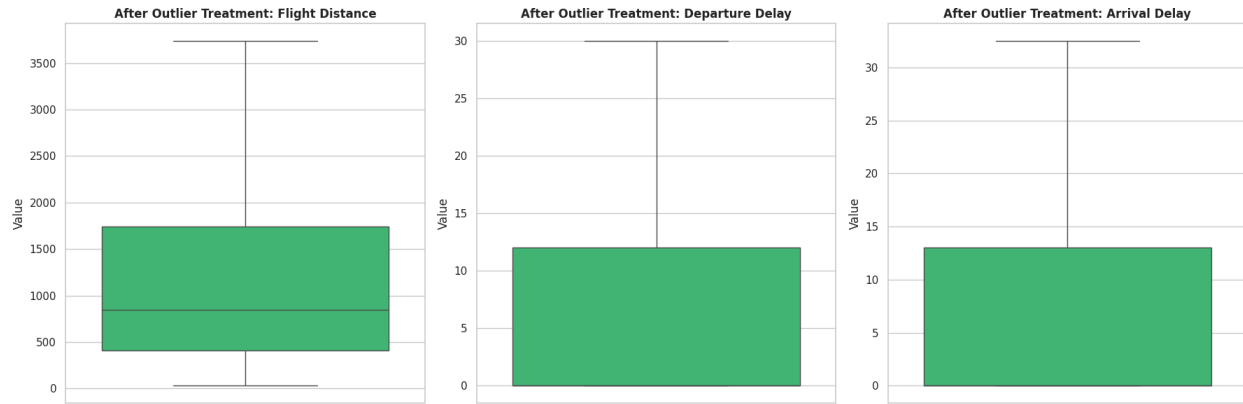


Figure 10: Outlier After Treatment

5.3. Feature Engineering

A cumulative travel disruption should be better represented by a Total Delay feature designed by summing delay in departure and arrival. Moreover, an Average Service Score was calculated as the sum of 14 attributes of rating related to the service, which gave a single measure of the general perceived service quality, and diminished dimensional redundancy (Kuhn and Johnson, 2013).

5.4. Categorical Encoding

Ordinal variables like Class to keep their natural order were coded using the Label Encoding. Nominal variables, such as Gender and Type of Travel, underwent One-Hot Encoding so as not to induce artificial ordinal associations, but in order to fit machine learning algorithms (Hastie et al., 2009).

5.5. Handling Class Imbalance (SMOTE)

The data did not have an extreme class imbalance (57 vs. 43). To resolve this, SMOTE was used on the training set only and was created by mixing synthetic minority-class using interpolation (Chawla et al., 2002). This led to an equal 50/50 distribution of the classes, which made the models more sensitive to contented passengers and ensured fairer performance ratings.

5.6. Data Quality Assessment

After detailed evaluation of data quality before training the model so that the reliability of the dataset could be ensured. It was found 99.7% of the records were complete and the missing values were limited to one feature. Duplicate records were recognized and eliminated, which was approximately less than 0.1% of the dataset. The consistency of data types was also checked throughout all the 24 features with no conflicting entries. The target variable distribution did not have any suspicious results or labeling problems that would affect the training of the model.

Cross-validation of service ratings (1-5 scale) ensured that there were no impossible values. The integrity of the entire dataset was good with only minor preprocessing involved. This strict quality measure assured the modeling performance would not be influenced by data artifacts and wrong measurements but true predictive trends.

6. Model Development

6.1. Train-Test Split Strategy

To make sure that both subsets were representative of the balanced population in terms of the class distribution, a stratified 80-20 train-test split was adopted. Models parameter estimates, and hyperparameter tests were performed using the training set (103,904 samples), and the test set (25,976 samples) was kept to the final, unbiased analysis of the performance metrics (Kuhn and Johnson, 2013).

6.2. Architecture and Configuration

All the models were trained on the same preprocessing pipeline, feature set, and train-test split to have a meaningful and fair comparison. Each of the models is a variant of a different learning paradigm: linear, bagging-based ensemble, and gradient boosting, which allows distinguishing between the differences in performance based on architectural features and not on how it handles the data.

6.2.1. Logistic Regression (Type: GLM):

A baseline classifier predictor was the Logistic Regression since it is simple, easily interpretable, and requires little calculation. It is a generalized linear model, which presupposes linear dependency between the input features and the log-odds of the target variable. Although this renders the model simple to explain and to train, it restricts its capacity to represent complicated non-linear interactions within the passenger satisfaction data.

- **Solver:** LBFGS
- **Activation Function:** Sigmoid
- **Regularization:** L2 ($C = 1.0$)
- **Maximum Iterations:** 1000

6.2.2. Random Forest (Type: Bagging Ensemble):

Random Forest was used as an ensemble model based on bagging which is meant to be able to capture non-linear relationships as well as being resistant to noises and outliers. The model is also able to train various decision trees on bootstrapped samples and combine their predictions with majority voting to reduce variance and achieve better generalisation. This renders Random

Forest quite appropriate to heterogeneous survey data consisting of ordinal service rating and correlated features.

- **Number of Trees:** 150
Split Criterion: Gini Impurity
- **Maximum Tree Depth:** 8
Minimum Samples per Leaf: 25
- **Feature Sampling:** Square root of total features
Aggregation Method: Majority voting

6.2.3. LightGBM (Type: Gradient Boosting Ensemble):

LightGBM was initialised as an optimised gradient boosting framework that is optimally adjusted to learn on large tabular data. In contrast to bagging methods, LightGBM creates trees one after another, where the trees are set to correct the errors of the preceding trees. Strong regularisation and structural constraints were used to ensure stable learning even with the high representational capacity of the model in case of overfitting and overly complex models.

- **Boosting Iterations:** 250
- **Learning Rate:** 0.008
- **Maximum Depth:** 3
- **Number of Leaves:** 15
- **Minimum Child Samples:** 80
- **Subsampling:** Row = 0.5, Column = 0.5
- **Regularization:** L1 (0.5), L2 (1.0)

7. Evaluation Metrics & Results

The models were tested on the unseen test set (20% of data). We put more emphasis on the metrics that are balanced in regard to precision and recall.

7.1. Comparative Performance Analysis

Accuracy, Precision, Recall, F1-Score and ROC-AUC were used to evaluate the models. Findings indicate that methods based on ensembles are far better than the linear baseline (Fawcett, 2006). The overall performance of the three models demonstrated that the Random Forest classifier was the most successful, with F1-score = 0.9349 and ROC-AUC = 0.9849 ranked higher than the Logistic Regression classifier (F1 = 0.8722) and LightGBM classifier (F1 = 0.9195). Additionally, PR-AUC scores recorded as: Random Forest = 0.9826, LightGBM = 0.9688, and Logistic Regression = 0.9330.

This shows that the interactions among the features of passenger satisfaction are nonlinear and complex, and they are better represented by the bagging-based ensembles. LightGBM still showed good results, but the generalisation capability and a better precision-recall trade-off in the test set was observed with Random Forest.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Time (s)	Prediction Time (s)
0	Random Forest	0.934978	0.934948	0.934978	0.934958	0.984900	0.982606	11.327846	0.436065
1	LightGBM	0.919503	0.919494	0.919503	0.919498	0.973876	0.968804	4.184580	0.369802
2	Logistic Regression	0.872113	0.872486	0.872113	0.872242	0.930460	0.933077	1.189313	0.005980

Figure 11: Model Performance Analysis

7.2. Confusion Matrix Analysis

The confusion matrix gives a detailed look at the errors:

- The confusion matrix underlines variation in error distribution between models. The number of misclassifications is highest in **Logistic Regression**, which is characterized by the general lack of the capacity to detect non-linear relationships.
- **Random Forest** has the most equalized performance, the maximum true positives and true negatives and minimum false positives and false negatives. This error profile is balanced which explains its best F1-score (0.9350) and shows reliable identification of both satisfied and dissatisfied passengers.
- **LightGBM** is competitive, yet has a higher number of false positives and false negatives as compared to Random Forest indicating a slightly weaker performance in borderline cases. In sum, the confusion matrix analysis proves that Random Forest is the best model to be deployed in operations.

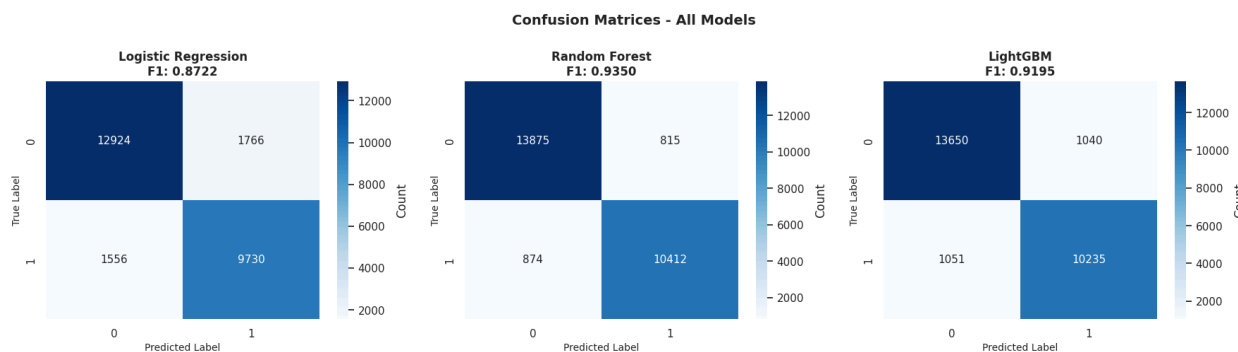


Figure 12: Confusion Matrix

Observation: According to the confusion matrix, Random Forest has the most correct predictions with the lowest number of false positives and false negatives which makes the

precisionrecall trade-off balanced. The misclassification rates are larger in the Logistic Regression and LightGBM is also competitive yet weaker, compared to the random forest.

7.3. Feature Importance

Feature importance analysis shows what the models actually learned.

- **Top Predictor:** The most important factor was online boarding. This is an indication that the initial digital touchpoint is a preview of the overall customer experience.
- **Second Predictor:** In-flight Wifi service. Connectivity is a necessity and not a luxury in the modern interconnected world.
- **Third Predictor:** Travel Class and Type. The satisfaction baseline of business travelers in premiere cabins is much higher, presumably because of superior amenities (Breiman, 2001).

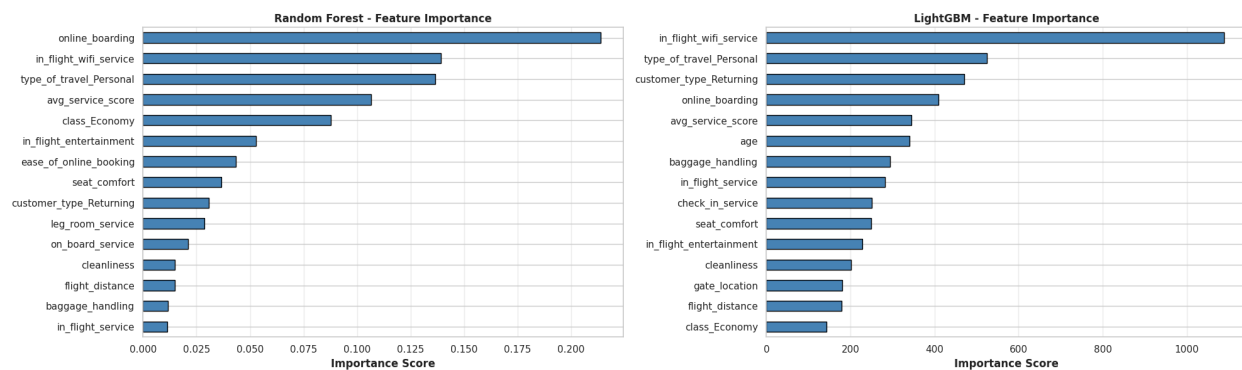


Figure 13: Feature Importance

Observation: The analysis of the feature importance demonstrates that Online Boarding and In-flight Wi-Fi Service have the most significant impact on passenger satisfaction, especially in the Random Forest model. The attributes of service quality and variables of travel context are at the top of the rankings, whereas operational characteristics, including the distance of the flight and flight delays, are less significant. This implies that it is the service experience but not operational metrics that pass on passenger satisfaction.

7.4. Permutation Importance Analysis

Permutation Importance was applied to give a model-independent measure of feature relevance in the trained LightGBM model (Breiman, 2001; Altmann et al., 2010). The approach scales how much performance (F1-score) of the model decreases when the performance is randomly permuted with the individual features, and the higher the performance decreases, the more significant the feature is.

Findings indicate that the Online Boarding influenced most the predictive accuracy, then In-flight Wi-Fi and In-flight Entertainment, which validates the prevalence of digital service quality in consumer satisfaction. Delay characteristics were rather moderate, which means that delays affect satisfaction but are not as significant as service experience. Generally, the permutation importance confirmed the strength and understandability of the model predictions, which proves its appropriateness in practical implementation (Altmann et al., 2010).

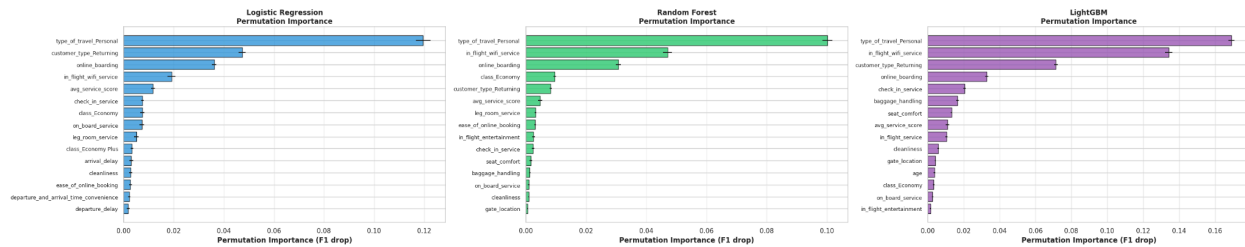


Figure 14: Permutation Importance

7.5. Learning Curve Analysis

A series of learning curves were created to determine the performance of the model with respect to the size of the training dataset. The curves can be used to give forecasts of convergence behaviour, generalisation-capacity, and the bias-variance trade-off by comparing training and validation scores to subsets of data of successively larger size. The analysis can be used to determine whether the performance boundaries are due to underfitting, overfitting or lack of training data (Kuhn and Johnson, 2013). A small difference between training and validation scores implies that generalisation is stable and further increase in validation performance implies that the model is using new data. On the whole, the learning curve analysis aids in evaluating the complexity of the model and makes sure the chosen model is appropriate in the prediction task (Hastie et al., 2009).

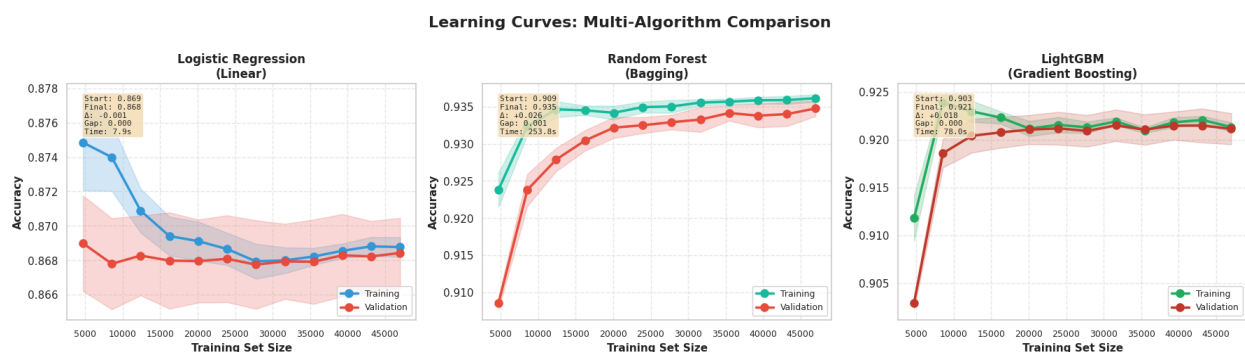


Figure 15: Learning Curve

Observation: Learning curves indicate that the performance is rising as the training data increases and a reduction in the gap between the training and validation scores. This shows good generalisation, low overfitting and correct model complexity.

7.6. Interpretation of Results

The relative analysis shows the evident performance deviation between the three models. Random Forest has turned out to be the most effective model by the highest F1-score (0.9350) and ROC-AUC (0.9849) and PR-AUC (0.9826) with a balanced precision recall trade-off. The model managed to achieve 0.93 or 93% accuracy with precision and recall of 0.93. Conversely, Logistic Regression was poor because it failed to capture any non-linear relationships and although LightGBM was also competitive, it did not pass the test set as compared to the performance of the Random Forest.

The findings show that the complex and non-linear relationships between service quality, travel context, and operational factors determine passenger satisfaction. Random Forest is a useful model in that it can capture these interactions well using ensemble aggregation, so it is especially resistant to noise, ordinal service ratings, and correlated features. That its false positives and false negatives are balanced further go in its favor as far as its appropriateness in operational decision-making is concerned.

The use of SMOTE also added to the better recall in the ensemble models as the identification of the satisfied passengers was dependable and was not biased towards the majority category. All in all, it can be concluded that bagging-based ensemble techniques offer a good compromise of precision, stability, and explainability such that Random Forest would be the best selection in predicting airline passenger satisfaction in this project.

8. Conclusion

8.1. Summary

The research established a reliable and replicable machine learning pipeline to predict airline passenger satisfaction based on large scale survey data. The random forest model is the most appropriate model through systematic preprocessing of data and feature engineering along with comparison analysis of different classifiers that showed good and balanced results with credible generalisation of unknown data.

The results prove that passenger satisfaction is more dependent on the service quality variables especially Online Boarding and In-flight Wi-Fi than on operational delays only. The degree of satisfaction also differed between passenger groups, with business-class and business-travel groups indicating an increase in satisfaction. Also, effects of delay were observed to be

non-linear and only significant effects were observed beyond certain thresholds, which supports the usefulness of non-linear, ensemble-based models in this prediction task.

8.2. Limitations

Although the methodology used in this study is strong, there are some key limitations that have to be mentioned. To begin with, the data is a time-stamped one and might not reflect changing passenger tastes, especially after the end of the pandemic travel trends, which will require periodic retraining of the model. Second, the source of undocumented data restricts the ability to generalize to other airlines that may offer different service quality levels or customer backgrounds. Third, there are no significant contextual variables, such as the price of tickets, competitiveness of routes, and causes of delays in particular, which can lead to unexplainable variance. Fourth, although SMOTE was able to balance classes, synthetic oversampling can introduce artificial patterns that are not found in real passenger populations. The fifth, the models were manually programmed with hyperparameters that were not optimized systematically, and so, performance improvements were not achieved. Lastly, no formal auditing of fairness along demographic subgroups was undertaken and this is essential to ethical use in customer-facing applications.

8.3. Reflection on Individual Learning

The significance of data preparation was also mentioned in this project, as it shows that preprocessing decisions are sometimes as important as model selection. It supported the requirement of proper measures in dealing with imbalanced data and demonstrated the superiority of ensemble models over linear methods in the case of nonlinear and conditional relations. The research also enhanced the knowledge on the contemporary gradient boosting methods, and their effectiveness benefits.

8.4. Future Recommendations

Random Forest is suggested to be deployed because of its level-balanced performance, strength, and interpretability. Further research should be aimed at monitoring models and retraining the workers on a regular basis to capture the changing behavior of the passengers and service standards. Operational value can further be enhanced by threshold optimization on the basis of business costs (e.g., putting emphasis on recall to lower customer churn). Predictive accuracy may be improved by segment-specific modelling of various travel classes or types of passengers. The further enhancements might be explainable AI methods like SHAP that will assist in making transparent decisions, predicting satisfaction in real-time, and examining advanced models on structured tabular data.

References

- Altmann, A., Toloşi, L., Sander, O., et al. (2010) Permutation importance: a corrected feature importance measure. *Bioinformatics*, 26 (10): 1340–1347. doi:10.1093/bioinformatics/btq134.
- Breiman, L. (2001) Random Forests. *Machine Learning*, 45 (1): 5–32. doi:10.1023/A:1010933404324.
- Chawla, N.V., Bowyer, K.W., Hall, L.O., et al. (2002) SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16: 321–357. doi:10.1613/jair.953.
- Dormann, C.F., Elith, J., Bacher, S., et al. (2013) Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, 36 (1): 27–46. doi:10.1111/j.1600-0587.2012.07348.x.
- Fawcett, T. (2006) An introduction to ROC analysis. *Pattern Recognition Letters*, 27 (8): 861–874. doi:10.1016/j.patrec.2005.10.010.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009) *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY: Springer New York. doi:10.1007/978-0-387-84858-7.
- Hess, S. and Polak, J.W. (2005) Mixed logit modelling of airport choice in multi-airport regions. *Journal of Air Transport Management*, 11 (2): 59–68. doi:10.1016/j.jairtraman.2004.09.001.
- Hosmer, D.W., Lemeshow, S. and Sturdivant, R.X. (2013) *Applied Logistic Regression*. Wiley Series in Probability and Statistics. 1st edn. Wiley. doi:10.1002/9781118548387.
- Ke, G., Meng, Q., Finley, T., et al. (2017) “LightGBM: A Highly Efficient Gradient Boosting Decision Tree.” In *Advances in Neural Information Processing Systems*. 2017. Curran Associates, Inc. Available at: https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html (Accessed: 31 January 2026).
- Kuhn, M. and Johnson, K. (2013) *Applied Predictive Modeling*. New York, NY: Springer New York. doi:10.1007/978-1-4614-6849-3.
- Park, J.-W., Robertson, R. and Wu, C.-L. (2004) The effect of airline service quality on passengers’ behavioural intentions: a Korean case study. *Journal of Air Transport Management*, 10 (6): 435–439. doi:10.1016/j.jairtraman.2004.06.001.
- Provost, F. and Fawcett, T. (2013) *Data science for business: what you need to know about data mining and data-analytic thinking*. First edition. Beijing Cambridge Farnham Köln Sebastopol Tokyo: O’Reilly.

Saha, G.C. and Theingi (2009) Service quality, satisfaction, and behavioural intentions: A study of low-cost airline carriers in Thailand. *Managing Service Quality: An International Journal*, 19 (3): 350–372. doi:10.1108/09604520910955348.

Tukey, J.W. (1977) *Exploratory data analysis*. Addison-Wesley series in behavioral science quantitative methods. Repr. Reading, Mass.: Addison-Wesley.

Zeithaml, V.A., Berry, L.L. and Parasuraman, A. (1996) The Behavioral Consequences of Service Quality. *Journal of Marketing*, 60 (2): 31–46. doi:10.1177/002224299606000203.